Should a milk tea shop add one temporary staff during peak hours?

A combined Monte Carlo and queuing risk analysis

Group A

Course: Simulation and Risk Analysis

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1 Introduction

This project is based on a real-life operating problem from a milk tea shop called Guozi's Tea in Tianjin. During the May Day holiday, one member of our project group visited this shop. Guozi's Tea is owned by Teacher Guo, who use to be a local comedy influencer. To attract more customer and increase store traffic, the shop was running a daily "second cup half-price" promotion. The shop has a clear peak period from 2 pm to 5 pm, and the promotion further increase customer demand and multi-cup orders. If customer arrivals and multi-cup orders increase faster than the shop's service capacity, the promotion may lead to longer waiting times, queue congestion, customer abandonment, and potential losses in customer experience. In this sense, a simple promotion becomes a capacity decision problem.

Based on this business problem, the project group design four possible operating scenarios for simulation.

Scenario A: Baseline represents normal operation with no promotion and two staffs.

Scenario B: Promotion Only represents the "second cup half-price" promotion with the original two staff members.

Scenario C: Add Capacity represents the promotion with one additional part-time workers during the three peak hours.

Scenario D: Improve Process represents the promotion with two staff members and workflow optimization, such as menu standardize, batching similar orders, and simplify the ordering and serving process.

These four scenarios allow the project to compare the impact of promotion, staffing decisions, and process improvements.

To answer the core question, this project uses a combination of **Monte Carlo Simulation** and **Discrete-Event Queueing Simulation** to evaluate how different operating strategies perform under uncertain customer demand, order patterns, service times, waiting times, and customer abandonment. The scenarios are compared based on five categories of key performance indicators: order and transaction volume, financial metrics, customer experience and queuing

performance, staff and operational efficiency, and risk. The goals of this project are to identify the most practical strategy that balances profits, service quality, and operational risk for Guozi's Tea during the 2–5 pm promotion peak period.

2 Parameters and Model Design

In this section, we would convert previous business issue into the simulatable, comparable and explicable model inputs.

2.1 Core Operating Assumptions

We initially begin with the basic operating assumptions.

1. Peak period length:

   The model focuses on the 3-hour peak period as the analytic window. The peak period is selected because the promotional risks always occur in the high-demand period because of rapid arrival, accumulated orders and extreme utilization rate rather than all day.

2. Main iterations & Sensitivity iterations:

   Each main scenrio runs 10,000 iterations to ensure obtain the stable profits and service distribution. For the sensitivity analysis, each selected parameter combinaion runs 1,000 iterations in order to judge whether the key input changes have the impact on the result.

3. Hourly wage & Maximum order size:

   In order to furnish the uniform operating baseline of comparing the later promotion, staff and process efficiency issues, hourly wage setted up as 25 yuan and maximum order sieze which is 8 cups are given for later comparison.

4. Random seed:

   In order to acquire the reproducible result, we have also fixed the random seed as 42.

See Appendix A.

2.2 Drink Category Assumptions

The investigated bubble tea shop has a large amount of SKU. Therefore, we simply the menus into four categories: basic, fruit, topping and premuim to avoid the unnecessary complexity of the model where it keeps the difference of price range, cost ratio and preparation time.

For the preparation time, we utilize the triangular distribution because it works well with lacking numerous historical data. In this way, we can only obtain the minimum, most likely and maximum preparation time to illustrate the distribution rather than an assumption of complex distribution. Thus, this distribution can not only reflect the nondeterminacy, but also can avoid the simplification between preparation time and fully fixed number. Meanwhile, the price range and cost ratio adopt the interval setting to reflect the difference of sales and original cost among the same category of drinks.

See Appendix A.

2.3 Scenario Design

In order to obtain the comparable result, we design the following four operating scenarios. The difference among these scenarios is the potential strategy the management would adopt facing the promotional issue rather than the random adjusting parameters.

Scenario A refers to the baseline operating situation with no promotion and 2 staffs. Scenario B adds the promotion but do not add extra staff which represents the direct but risky strategy because of the demand-supply issue in the promotional side. Scenario C not only adds the promotion but also adds 1 extra staff which could reduce the waiting time and lossing customer issues by right of the extra capacity. Scenario D keep the same staff amount, yet add the promotion and improve the process improvement such as handling large similar orders. In our model, scenario D reduces the preparation time multiplier and fix order time to represent this process improvement. In this case, scenario C and D is our potential strategies to solve the promotional issue.

See Appendix A.

2.4 Key Model Parameters

After the basic operating assumptions, drink category and scenario design, the last stage is to declare how these setting enter the ultimate simulation. In this section, the key parameters linking to the business issue would be explained where they mutually determine the direction of the change of profit, waiting time, staff utilization and customer loss in the later process. This is the summary of the key model parameters.

See Appendix A.

3 Model Design

3.1 Order Generation

We used the Monte Carlo simulation model and queuing model to simulate the operation of the tea shop during peak hours, so that the simulation results could be as close as possible to the real-world situation. In this way, the milk tea shop can analyze the simulation results and make valuable decisions.

3.1.1 Order Quantity

We calculate the average number of orders during peak hours.

$$\text{Expected Arrivals} = \lambda \times \text{Peak Hours} \tag{1}$$

Therefore, the actual order quantity follows a Poisson distribution.

$$N \sim \text{Poisson}(\text{Expected Arrivals}) \tag{2}$$

3.1.2 The Arrival Time of the Order

Next, we will calculate the arrival time for each order within the range of 0 to 180 minutes, and then sort the orders by time from earliest to latest.

For example:

$$[0.5,\ 1.8,\ 3.2,\ 5.0,\ \ldots,\ 179.4] \tag{3}$$

In this way, the queuing model can process each customer one by one according to their arrival order.

3.1.3 Definition of the Number of Cups per Individual Order

Through the research on milk tea shop, we found that the ordering system of these shops allows for a maximum of 8 cups per single order. Additionally, in real life, as the number of cups in an order increases, the probability of occurrence generally decreases. Therefore, when simulating the number of cups in an order, we adopted the truncated geometric distribution:

$$P(Q = q) = \frac{(1 - r)r^{q-1}}{1 - r^8}, \qquad q = 1, 2, \ldots, 8, \quad 0 < r < 1. \tag{4}$$

- When q increases, $P(Q = q)$ decreases.
- r determines the frequency of multiple-order requests.
- The larger the r value, the greater the possibility of a large order occurring.
- The maximum number of cups in an order is 8.

Since $0 < r < 1$, r^{q-1} will decrease as q increases. Therefore, we can achieve the result that "the more cups, the smaller the probability."

3.1.4 Definition of Individual Order Revenue

Through the survey of milk tea shop, we found that the charging rule for "half-price on the second cup" in the shop is as follows: First, sort the prices of the drinks in descending order. Group them into pairs of two cups each. The second cup with the lower price will be half-price; if there is an odd number of cups, the remaining one will be at the original price.

For example, for an order of 5 cups with prices being $[18, 15, 12, 8, 6]$, the half-price items are 15 and 8, and the revenue is:

$$18 + 15 \times 0.5 + 12 + 8 \times 0.5 + 6 = 47.5. \tag{5}$$

More generally, after sorting drink prices in descending order as $p_{(1)} \geq p_{(2)} \geq \cdots \geq p_{(Q)}$, the order revenue can be written as:

$$R = \sum_{k=1}^{Q} d_k p_{(k)}, \qquad d_k = \begin{cases} 1, & k \text{ is odd}, \\ 0.5, & k \text{ is even}. \end{cases} \tag{6}$$

3.1.5 Cost of the Order

The cost of each drink is the simulated price multiplied by a random cost rate:

$$\text{Cost} = \text{Price} \times \text{Cost Ratio}. \tag{7}$$

3.1.6 Service Time of the Order

One cup order:

$$\text{Service Time} = \text{Fixed Order Time} + 1 \times \text{Prep Time}. \tag{8}$$

Multiple cup order:

$$\text{Service Time} = \text{Fixed Order Time} + \text{Effective Batch Efficiency} \times \text{Prep Time Sum}. \tag{9}$$

Note: Effective Batch Efficiency is used to indicate that some steps in a multi-cup order can be shared, so the service time will not simply be equal to the sum of the individual preparation times for each cup.

Finally, these steps will result in the following outcomes:

See Appendix B.

3.2 Queuing Model

Based on the research of the milk tea shop and the observation of the actual situation, our queuing model should be a limited-duration $M/G/c$ queuing system with customer churn, implemented through discrete event simulation, that is:

- M: Customer arrivals follow a Poisson process (Markovian), with an arrival rate of λ people per unit time.
- G: The service time follows any general distribution (General), only requiring the existence of the mean $E[S]$ and variance $\mathrm{Var}(S)$.
- c: c parallel service counters ($c \geq 1$), customers queue in single groups and are served on a first-come, first-served basis (FCFS).

We process the orders one by one according to the order in which the customers arrive.

And our explanation for the capacity of the milk tea shop is: Based on the recollections of the milk tea shop staff, we set the patience level of the customers. If the waiting time exceeded the customers' patience level, then the customers would leave.

3.2.1 Queueing Logic

When a customer arrives, find an available employee or the one who finished work earlier to serve that customer.

$$j^* = \arg\min_{j=1,\ldots,c} T_j \tag{10}$$

$$E = T_{j^*} \tag{11}$$

where T_j is the next free time of server j, j^* is the selected server, and E is the earliest available time.

3.2.2 Customer Waiting Time

$$\text{Waiting Time} = \max(0,\ \text{Earliest Server Free Time} - \text{Arrival Time}). \tag{12}$$

If the staff is available when the customer arrives, namely the earliest available time is less than the arrival time, then the waiting time should be 0.

If the employee is not available, namely the earliest available time is greater than the arrival time, then the customer will have to wait:

$$\text{Waiting Time} = \text{Earliest Available Time} - \text{Arrival Time}. \tag{13}$$

3.2.3 Update Employee Status

Service start time:

$$\text{Service Start} = \max(\text{Arrival Time},\ \text{Earliest Available Time}). \tag{14}$$

Service end time:

$$\text{Service Finish} = \text{Service Start} + \text{Service Time}. \tag{15}$$

Then update the status of the employees:

$$\text{Server Free Time} = \text{Service Finish}. \tag{16}$$

The meaning is that after this employee takes on this order, they have to finish it before they can take on the next one.

3.2.4 The Busy Hours of the Employees

The significance of this calculation is due to the fact that some orders may start being served at the 179th minute and continue until the 183rd minute. However, our peak period only lasts from 0 to 180 minutes. Therefore, the employees were only busy for $180 - 179 = 1$ minute during the peak hours.

3.2.5 Total Profit

First, we determine whether the customer has left. If the waiting time exceeds the customer's patience, the customer will leave and the profit will be 0.

If Waiting Time $>$ Customer Patience, then the customer abandons, and:

$$\text{Revenue} = 0, \qquad \text{Variable Cost} = 0. \tag{17}$$

The lost potential profit is:

$$\text{Lost Potential Profit} = \text{Revenue} - \text{Variable Cost}. \tag{18}$$

If the customer does not leave, namely Waiting Time $<$ Customer Patience, then the profit will be:

$$\text{Profit} = \text{Revenue} - \text{Variable Cost} - \text{Labour Cost}. \tag{19}$$

The labour cost is:

$$\text{Labour Cost} = \text{Staff} \times \text{Hourly Wage} \times \text{Peak Hours}. \tag{20}$$

For example, there are 2 employees, with an hourly wage of 25, and they work for 3 hours during peak hours:

$$2 \times 25 \times 3 = 150. \tag{21}$$

3.2.6 Queueing Indicators

$$\text{Average Waiting Time} = \frac{\sum \text{Waiting Time of Served Orders}}{\text{Served Orders}}. \tag{22}$$

$$\text{Probability of Waiting, All} = \frac{\sum\{\text{Arrived Orders with Waiting Time} > 0\}}{\text{Total Arrivals}}. \tag{23}$$

$$\text{Probability of Waiting, Served} = \frac{\sum\{\text{Served Orders with Waiting Time} > 0\}}{\text{Served Orders}}. \tag{24}$$

$$\text{Probability of Waiting Over 8 Minutes, All} = \frac{\sum\{\text{Arrived Orders with Waiting Time} > 8\}}{\text{Total Arrivals}}. \tag{25}$$

$$\text{Probability of Waiting Over 8 Minutes, Served} = \frac{\sum\{\text{Served Orders with Waiting Time} > 8\}}{\text{Served Orders}}. \tag{26}$$

$$\text{Abandonment Rate} = \frac{\text{Lost Orders}}{\text{Total Arrivals}}. \tag{27}$$

$$\text{Average Queue Length} = \frac{\sum \text{Waiting Time}}{\text{Peak Minutes}}. \tag{28}$$

$$\text{Utilisation} = \frac{\text{Total Busy Time}}{\text{Staff} \times \text{Peak Minutes}}. \tag{29}$$

$$\text{Workload Ratio} = \frac{\text{Total Service Time}}{\text{Staff} \times \text{Peak Minutes}}. \tag{30}$$

Finally, all of the indicators and their explanation will result in the following outcomes.

See Appendix C.

Note: The formulas for some indicators are not provided in the original document, but the calculation logic is similar to that of other formulas.

Finally, following the above steps, we conducted 10,000 simulations for each of the 4 scenarios. We observed a result that is as close as possible to the real situation. This is significant for

helping the milk tea shop determine the appropriate strategy.

4 Promotion & Queue Risk Analysis and Sensitivity Analysis

4.1 Promotion & Queue Risk Analysis

4.1.1 Significance of Risk Analysis

Traditional performance evaluation relies heavily on average profit, but this single metric cannot capture the downside volatility inherent in retail operations. For the bubble tea shop, running a promotion activity attracts more customers and enhances revenue, but it also brings a trade-off: longer waiting times and higher customer abandonment rates may occur at the same time. These negative outcomes can erode the expected benefits of promotions and even result in net losses in extreme scenarios. Therefore, risk analysis is significant because it quantifies the risk in the two improved scenarios we set, enabling data-driven decision-making that balances profitability and stability.

4.1.2 Modeling Approach

First, we run the Monte Carlo simulation to model the inherent randomness in customer behavior and operational processes, with key inputs including customer arrival rate, order size, drink mix, preparation time, and customer patience. Second, the discrete-event queuing model is used to mimic the real-time service process and queuing dynamics in the store, incorporating inputs such as staff count, First-Come-First-Served (FCFS) service logic, and customer abandonment threshold. Finally, we conduct a risk analysis to quantify the downside financial risk of each strategy. The evaluation metrics include VaR_{95}, CVaR_{95}, and probability of missing target.

4.1.3 Risk Metrics Definition

We use the average profit of Scenario A, no promotion with two staff, as the baseline target profit, as it represents the store's normal operating performance without promotional interference. For each simulation run, we define the profit shortfall as the amount by which actual profit falls below this baseline:

$$\text{TargetProfit} = \text{MeanProfit}_A \tag{31}$$

$$\text{Shortfall} = \max(\text{TargetProfit} - \text{Actual Profit}, 0) \tag{32}$$

If the actual profit exceeds the baseline, the shortfall is zero, indicating no downside loss for that run. We also apply VaR_{95} and CVaR_{95} to measure tail risk:

$$\text{VaR}_{95} = 95\text{th percentile of the shortfall distribution} \tag{33}$$

VaR_{95} represents the threshold at which the worst 5% of outcomes begin, providing a clear measure of the maximum expected loss under normal operating conditions.

$$\text{CVaR}_{95} = E\left[\text{Shortfall} \mid \text{Shortfall} \geq \text{VaR}_{95}\right] \tag{34}$$

CVaR_{95} represents the average profit shortfall experienced in the worst 5% of simulated scenarios. In addition, we calculate the probability of missing target, which is the percentage of simulation runs where the strategy's profit falls below the baseline Scenario A profit.

Probability of Missing Target is defined as the statistical probability that actual output below the predefined target profit in scenario A.

$$P(\text{Missing Target}) = P(\text{Actual Value} < \text{Target value}_{\text{scenario A}}) \tag{35}$$

4.1.4 Risk Analysis Results

Downside Risk Metrics Across Four Strategies

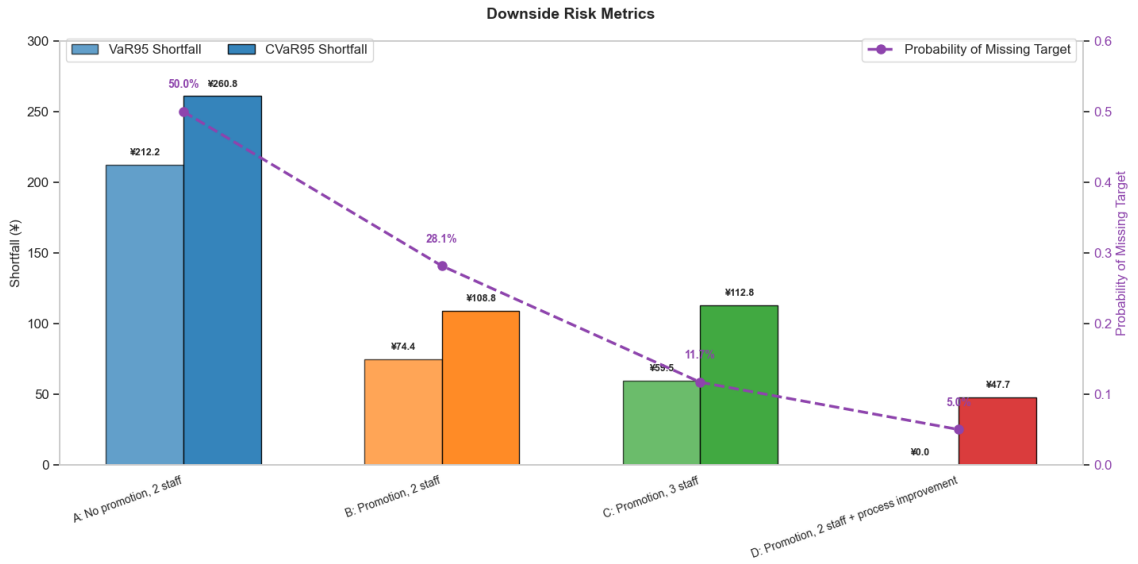


Figure 1: Downside risk metrics across four strategies.

The chart displays VaR_{95} shortfall, light color bars, CVaR_{95} shortfall, dark color bars, and the probability of missing target, purple dashed line with markers, for each scenario.

Summary of Downside Risk Metrics

See Appendix D.

Scenario A, no promotion with two staff, is the riskiest option, and it has a 50% probability of underperforming its own average with the highest CVaR_{95}. Scenario D, promotion with two staff and process improvement, is the lowest-risk strategy, achieving a VaR_{95} of 0. This demonstrates that process improvements can effectively eliminate downside risk while capturing the revenue benefits of promotions. However, although Scenario C, with an extra staff member but unimproved process, reduces the probability of missing target compared with Scenario B, its CVaR remains similar to Scenario B. This indicates that additional staff only reduce the frequency of losses rather than eliminating the severity of extreme tail events.

4.2 Sensitivity Analysis

4.2.1 Purpose

Sensitivity analysis identifies which parameters have the greatest impact on profit. By identifying the most influential factors, we can prioritize operational improvements, allocate resources effectively, and reduce risk.

4.2.2 Methodology

We conduct a one-at-a-time (OAT) sensitivity analysis using Scenario B, promotion with two staff, as the baseline. Scenario B is chosen because it represents the high-risk situation of running a promotion without increasing staff numbers or improving the process. It is more sensitive to changes in operational parameters, as the system is already operating near its capacity limit, making it ideal for identifying critical thresholds. The OAT approach adjusts one parameter to its minimum and maximum values while keeping the other variables unchanged. The sensitivity of profit to each parameter is measured by the absolute swing in mean profit:

$$\text{Swing} = |\text{MeanProfit}_{\text{high}} - \text{MeanProfit}_{\text{low}}| \tag{36}$$

In this way, a larger swing indicates a greater impact on profit, which means this parameter should be monitored more carefully in future operations.

4.2.3 Tested Parameters

We test five key operational and business parameters that drive the store's performance. The low and high settings for each parameter are based on historical data and realistic operational bounds.

We have two output graphs, one is without parameters normalization, while other is not. The reason to normalize the parameters in sensitivity analysis is that theses parameters usually have different units, value ranges, and magnitudes, and without normalization, parameters with larger absolute values may dominate the sensitivity results, leading to misleading comparisons of parameter importance, causing bias and misinterpretation of the output. Hence the normalization ensures we obtain the true impact of the parameter to the model.

See Appendix E.

4.2.4 Sensitivity Analysis Results

Tornado Chart: Profit Sensitivity Analysis

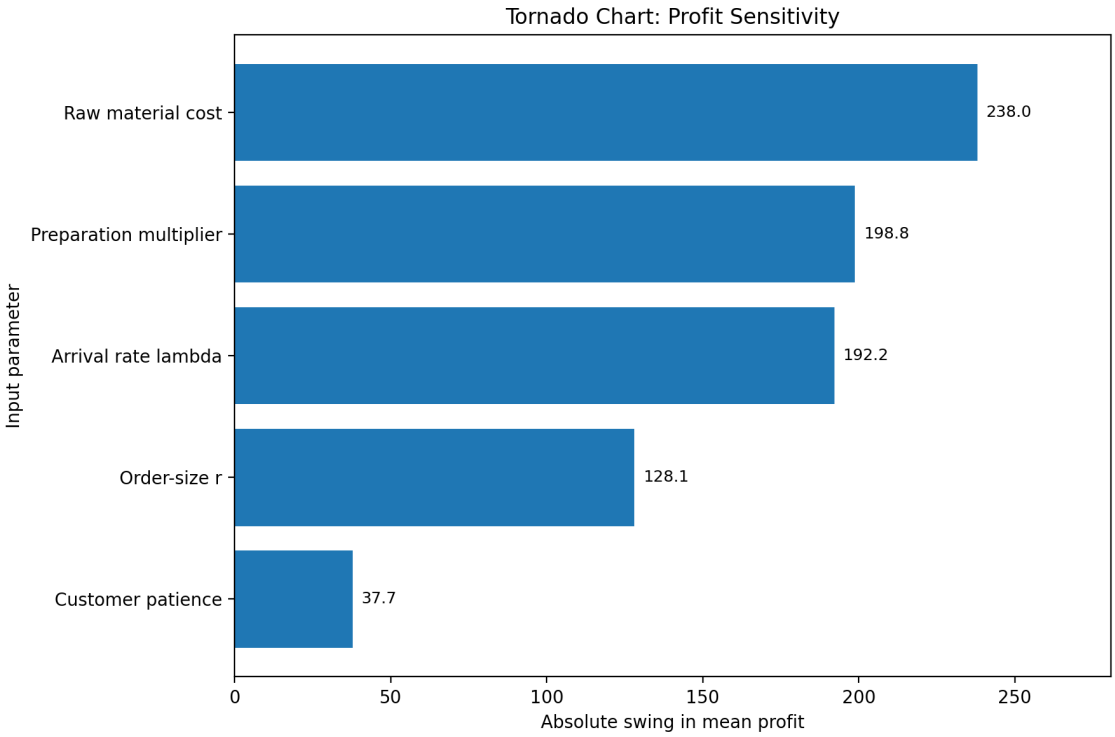


Figure 2: Tornado chart for profit sensitivity analysis.

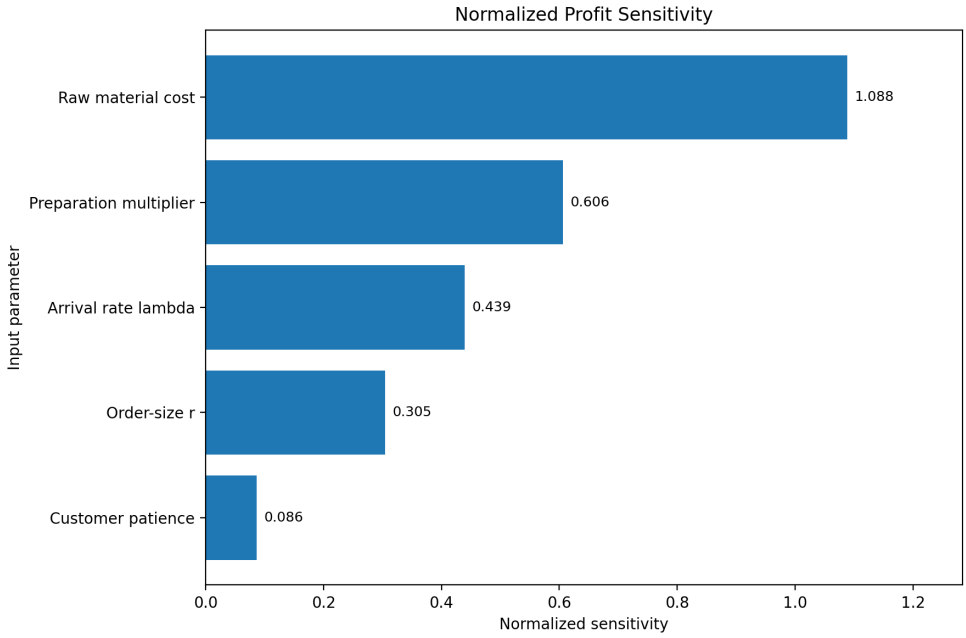


Figure 3: Tornado Chart: Profit Sensitivity

Comparing two tornado graphs, with and without parameter normalization, they show the same rank in their output. The sensitivity analysis results clearly rank the impact of each parameter on profit. Raw material cost is the most influential factor on profit; preparation time multiplier is the second most important factor; customer arrival rate has a significant impact on profit; order-size parameter has a moderate impact; and customer patience has the least impact on profit.

5 Results and Analysis

This section compares four business scenarios based on the results of Monte Carlo simulation and queueing simulation.

The analysis focuses not only on average profit but also on waiting time, customer churn rate, queue length, and employee utilization rate because the core business question of this study is: When a milk tea shop offers a **second-drink-at-half-price** promotion, should it prioritize adding one more staff directly or optimize processes to reduce queue pressure, thereby enhancing both profits and the customer experience.

5.1 Mean Profit

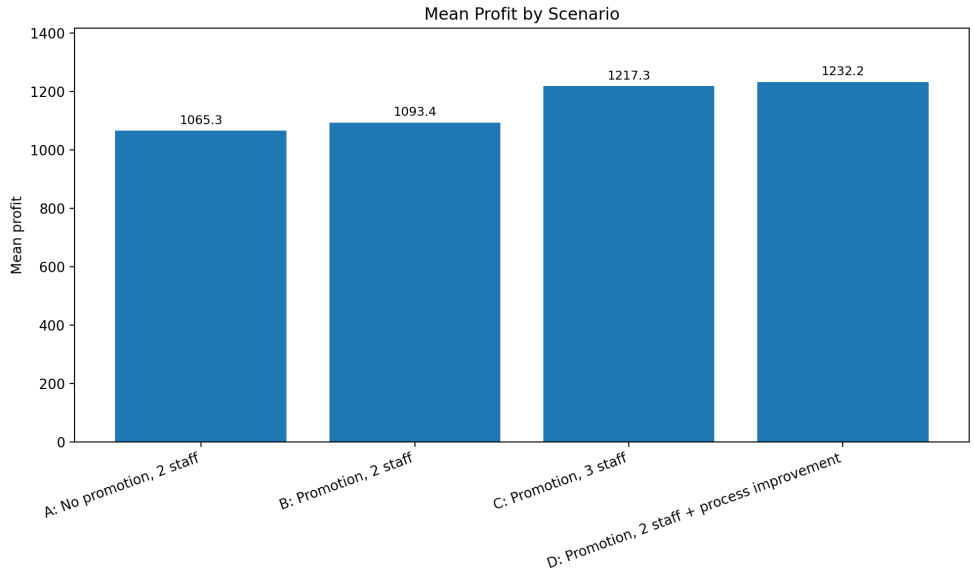


Figure 4: Mean Profit by Scenarios

As can be seen from the average profit graph, the promotion itself does bring higher revenue, since the profit of the three promotion scenarios B, C, and D is higher than non-promotion

scenario A. Among them, D has the highest profit, indicating that in the case of existing promotions bringing about increased customer flow, the improvement of process efficiency can convert more potential demand into actual profits. C has a slightly lower profit than D, mainly due to the added cost of additional labor.

5.2 Average Waiting Time

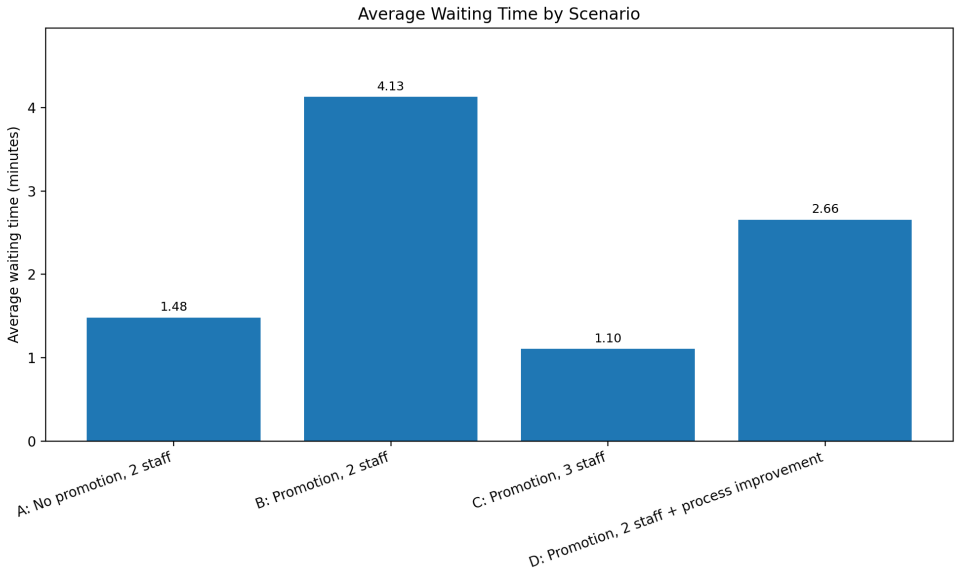


Figure 5: Average Waiting Time by Scenario

The waiting-time chart shows that Scenario B has the longest waiting times among the four options. This indicates that the additional customers attracted by the promotional campaign could not be effectively served by the existing two staff members, resulting in longer waiting times for customers. For businesses like milk tea shops, where demand peaks during busy periods and customers make decisions quickly, excessive waiting times often directly affect conversion rates and the likelihood of repeat purchases.

By contrast, Scenario C has the shortest waiting times, indicating that increasing the number of staff members is the most direct and effective way to alleviate congestion. Although Scenario D is not as good as Scenario C, it is still significantly better than Scenario B, showing that process optimization can also help reduce waiting times, albeit to a lesser extent than simply adding more staff.

5.3 Customer Abandonment

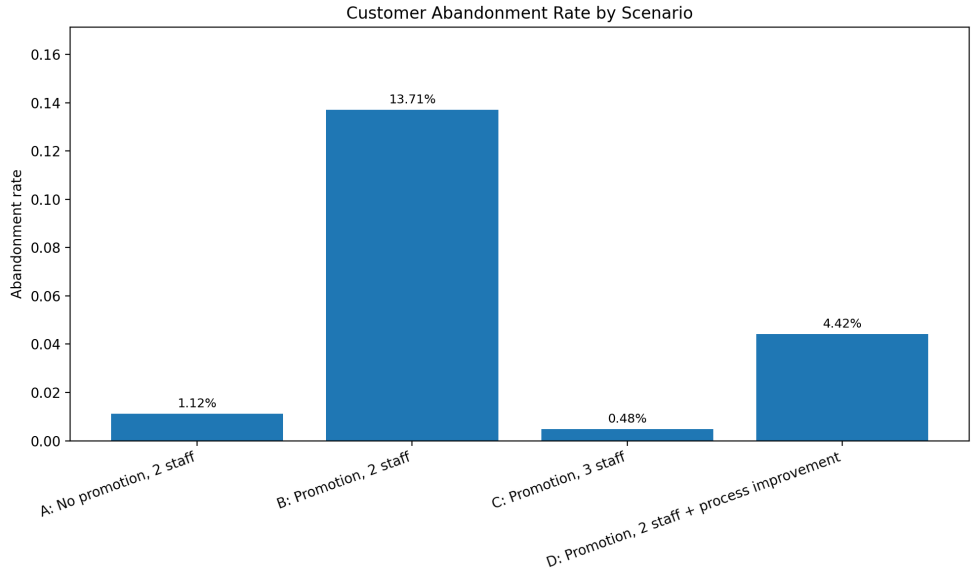


Figure 6: Customer Abandonment Rate by Scenario

Abandonment rate results are consistent with wait time trends. Scenario B has the highest rate of nearly 14%, meaning that on average, about 14 out of every 100 in-store orders abandon because they wait too long. For promotional activities, this is a typical **demand-but-can't-catch** situation, which also means that marketing expenses and brand exposure are not fully converted into sales revenue.

In contrast, Scenario C's abandonment rate is almost negligible at 0.48%; D also reduced the abandonment rate to 4.42%. This shows that as long as the store is willing to make adjustments on operating capacity, either by increasing staff or optimizing processes, it can significantly reduce the demand loss caused by queuing.

5.4 Staff Utilisation and Average Queue Length

See Appendix F.

The result shows the average queue length and employee utilization rates under four different scenarios. These two indicators can more directly reflect whether the system is approaching saturation.

In terms of average queue length, **Scenario B** has the highest value, reaching 2.97. This indicates that conducting promotional activities without increasing service capacity will lead to longer

queues during peak hours. In contrast, **Scenario A and Scenario C** have average queue lengths of around 1.05 and 1.04, respectively, suggesting that the baseline scenario without promotions and the "promotion plus one additional employee" scenario are both able to effectively control queue sizes. **Scenario D** has an average queue length of 2.13, which is still higher than that of Scenarios A and C but significantly lower than Scenario B, indicating that process optimization can help alleviate congestion issues without adding more staff.

Regarding employee utilization rates, **Scenario B** also has the highest rate, at 94.41%. This means that the system is operating very close to its full capacity, with little buffer space left, making it more prone to service delays and customer loss. In comparison, **Scenario A and Scenario C** have employee utilization rates of 74.49% and 74.32%, respectively, indicating that these two scenarios allow for a sufficient margin of service capacity. The employee utilization rate in **Scenario D** is 87.32%, which falls between that of Scenario B and Scenario A/C. This indicates that process optimization has improved system efficiency; however, the stores are still under a high level of load.

5.5 Risk and Downside Analysis

See Appendix G.

From the results, Scenario D has the least downside risk. Its VaR95 shortfall is 0.00, and its CVaR95 shortfall is the lowest, at only 47.67, indicating its most stable profit performance under adverse conditions. In other words, in the vast majority of simulation scenarios, D's profit has reached or exceeded the benchmark level, and even in the worst-case tail scenario, its average shortfall is significantly lower than other scenarios.

In comparison, Scenario B's VaR95 shortfall is 74.36, and CVaR95's is 108.81; Scenario C's VaR95 shortfall is 59.51, CVaR95 is 112.78. This indicates that in the worse scenario, the profit volatility of both schemes is greater than that of D. Although C's average profit is higher than B's, due to increased labor costs, it does not outperform D in tail risk.

From a business perspective, process optimization not only increases average profit, but also reduces profit losses in extremely adverse situations. Therefore, from a risk control perspective, Scenario D is a more robust solution.

5.6 Sensitivity Analysis

Figure 7 shows the results of the sensitivity analysis. The horizontal axis represents `swing_abs_profit`, which indicates the absolute change in average profit when a particular parameter is changed from the low setting to the high setting. The larger the value, the more sensitive that parameter is to the profit outcome.

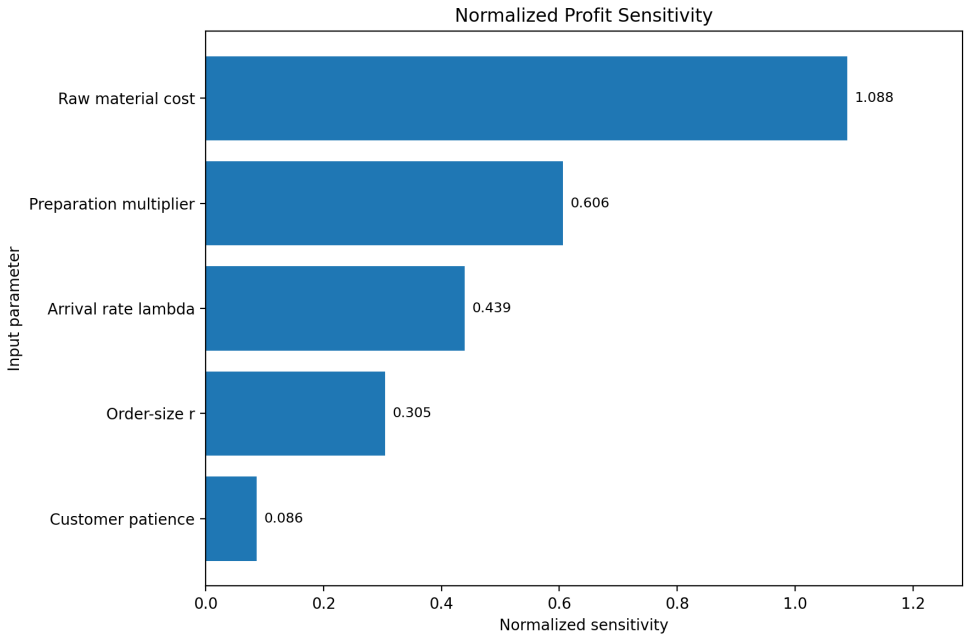


Figure 7: Tornado Chart: Profit Sensitivity

Sensitivity analysis indicates that the three factors that have the most significant impact on profits are: **Raw material Cost**, **Production Speed**, and **Customer Arrival Rate**.

Among them, the arrival rate reflects whether the promotion can really bring enough passenger flow; The cost of raw materials directly affects the marginal profit of each order; The production speed determines whether the store can convert peak demand into actual transactions. In contrast, customer patience, while affecting churn rates, has a relatively small overall impact on the bottom line.

This has two direct implications for management practice:

First, when doing promotions, stores should not only pay attention to "whether more customers are coming", but also pay attention to whether the supply chain and on-site execution capabilities

can keep up.

Second, if the store wants to improve results without significantly increasing labor costs, prioritizing improving cup dispensing efficiency will be more patient and more effective than simply hoping for customers.

5.7 Business Interpretation

In conjunction with on the interpretations derived from the results of the previous four sections, the main commercial implications of this study can be summarized in the following three points.

Firstly, promotions can indeed increase profits, but they also significantly increase waiting times. Compared to baseline Scenario A, Scenario B resulted in an average profit increase from 1065.28 to 1093.42, indicating that the "second drink at half price" promotion effectively boosted sales and revenue. However, at the same time, the average waiting time in Scenario B increased to 4.13 minutes, the customer churn rate reached 13.71%, and employee utilization rates were as high as 94.41%. This shows that promotional activities are not cost-free; they put systems under near-full-load conditions. If stores focus solely on revenue growth while neglecting service capacity, some of the potential profits will be lost due to increased waiting times and customer attrition.

Secondly, adding an additional employee can significantly improve service quality. Under promotional conditions, Scenario C reduced the average waiting time to 1.10 minutes and further decreased the customer churn rate to 0.48%, demonstrating that increasing staff numbers is highly effective in alleviating congestion.Although the labor costs associated with option C are higher, its average profit is still greater than that of option B. This indicates that the additional human resources invested have not been wasted, but rather have yielded benefits by reducing queue delays and enhancing the stability of service. In terms of results, increasing the number of employees is a more direct and reliable way to alleviate congestion.

Thirdly, profit risks are primarily driven by factors such as demand, costs, and production efficiency. Sensitivity analysis reveals that the three factors that have the greatest impact on average profit are customer arrival rates, raw material costs, and production times. This means

that whether a store can actually benefit from promotional activities depends not only on whether its marketing efforts attract enough customers but also on whether raw material costs are under control, as well as on how quickly orders can be processed in both front and back office. In comparison, although customer patience can affect the rate of customer turnover, its overall impact on profits is relatively minor.

Overall, these results indicate that promotional strategies cannot be evaluated in isolation from a store's operational capabilities. For businesses like milk tea shops that operate during peak hours, increased demand can only be effectively translated into stable profits if there are sufficient staff members or if process efficiency is improved; otherwise, additional customer traffic may lead to longer waiting times and higher turnover rates.

6 Final Conclusions and Recommendations

6.1 Final Evaluation of Scenarios

According to our results, Scenario D has the highest average profit at $1,232.2. Although it did not increase the number of employees during the promotion, it turned the customer churn rate from 13.71% in Scenario B to 4.42% and reduced average waiting time from 4.13 minutes to 2.66 minutes. This was achieved by optimizing the operation process. More importantly, Scenario D produced the lowest downside risk among all scenarios. Its VaR95 shortfall was 0 and the CVaR95 shortfall was 47.67. This shows that the profit outcomes are more stable comparing with the other scenarios.

From a management perspective, Scenario D is the most recommended strategy for the current model. This indicates that it doesn't necessarily have to include additional workers during promotion; instead, it focuses on raising operational efficiency.

In practice, this could include:

1. Simplifying promotion menus
2. Batching similar drink orders
3. Preparing popular ingredients in advance

4. Reducing highly customized requests

For high-demand drinks, stores may also create a separate fast-track preparation flowline to simplify the operation and shorten the time for serving.

Scenario D depends heavily on whether the process of producing and ordering is optimized. However, in practice, if the store already has a highly standardized operating process, there is limited room for optimizing. In that case, Scenario C would be a safer alternative.

Although Scenario C has a slightly lower average profit than Scenario D, it delivers the best service performance, has the lowest average waiting time of 1.1 minutes and customer churn rate, while also maintaining lower staff utilization of 74.32% and shorter queue lengths. In comparison, Scenario D recorded a waiting time of 2.66 minutes and a higher utilization level of 87.32%.

Scenario C provided a larger operational buffer acting as the safety net, reducing the risk of system congestion during peak hours of demand. Lower staff utilization also means employees were less likely to operate near full capacity, which helped maintain shorter queues, lower customer churn and higher future customer loyalty. This shows that adding one temporary employee can quite effectively improve service stability when customer traffic becomes volatile. Therefore, even if Scenario D has the highest profitability, when the demand is volatile or there is limited room for process optimization, Scenario C presents a strategy that is stable and reliable. More importantly, the current excellent customer experience may be transformed to future excellent reputation and higher customer loyalty.

Overall, Scenario D is the profit-maximizing strategy, and Scenario C is the stable-service-oriented strategy. The final operational decision should depend on whether there is room for optimizing realistic processes within the store.

6.2 Recommended Strategy

Scenario D is the most recommended strategy. According to the simulation, this involves maintaining operations with two employees during the promotion while simultaneously introducing workflow optimization measures. Suitable actions may include limiting the range of drinks

included in promotions, standardizing high-frequency drinks into simplified promotional menus, reducing complicated customization requests, and preparing key ingredients for popular drinks in advance. For products with extremely high demand, we will design a separate flow line for rapid preparation to reduce service time.

Scenario C should also be considered as the backup strategy. If the process optimization cannot be implemented in practice, or if the actual customer stream is much higher than expectation, we should add one temporary employee.

In contrast, Scenario B is not recommended as a standalone strategy. Although promotions alone may increase the average profit rate, the failure to increase the capacity for service or improve the efficiency of the workflow may lead to longer queues and higher customer churn. This may negatively affect customer experience and brand reputation.

Finally, based on the sensitivity analysis, managers should not only attract customers but also ensure the supply chain and on-site operational capabilities. In addition, under the premise of controlling labor costs, improving the efficiency of serving customers is a more reliable and effective approach than simply waiting for customers to come.

7 Model Limitations

This simulation model still has some limitations. First of all, the price of the drink, production costs, and preparation times were based on the brief description of the merchant instead of real operational data distribution collected from the stores.

Second, customer patience was approximated using a triangular distribution, while in reality this may be influenced by factors like queue length, weather conditions, store reputation, and customer expectations.

Third, the model assumes that one worker handles one order at a time without separating different operational tasks such as: cashiering, sealing, delivery and customer communication.

Therefore, the effectiveness of the process optimization in Scenario D also requires real-world validation, since the actual improvement in preparation speed may vary across stores.

Furthermore, production costs were allowed to fluctuate randomly within drinking categories. If real menu data were available, future models could assign fixed prices and fixed costs to individual drinks.

Therefore, the simulation model should be viewed as a tool for comparing the performance of different operational strategies, rather than predicting the exact profit of a specific milk tea store.

8 References

References

[1] Batt, R. J., & Terwiesch, C. (2015). Waiting patiently: An empirical study of queue abandonment in an emergency department. *Management Science, 61*(1), 39–59. https://doi.org/10.1287/mnsc.2014.2058

[2] Dai, J. G., & He, S. (2011). Queues in service systems: Customer abandonment and diffusion approximations. *Tutorials in Operations Research*, 36–59. https://doi.org/10.1287/educ.1110.0083

[3] Green, L. V. (2011). Queueing theory and modeling. In Y. Yih (Ed.), *Handbook of healthcare delivery systems*. Taylor & Francis. https://business.columbia.edu/faculty/research/queueing-theory-and-modeling

[4] Gopalakrishnan, R., & Zhong, Y. (2023). *Some asymptotic properties of the Erlang-C formula in many-server limiting regimes*. arXiv. https://arxiv.org/abs/2304.13845

[5] Kambli, A., Sinha, A. A., & Srinivas, S. (2020). Improving campus dining operations using capacity and queue management: A simulation-based case study. *Journal of Hospitality and Tourism Management, 43*, 62–70. https://doi.org/10.1016/j.jhtm.2020.02.008

9 Appendix

A Key Model Parameters

Table A.1: Core Simulation Setup

Parameter	Setting
Peak period length	3 hours
Main iterations	10,000 per scenario
Sensitivity iterations	1,000 per low/high case
Hourly wage	25
Maximum order size	8 cups
Random seed	42

Table A.2: Drink Category Assumptions

Category	Price range	Cost ratio	Preparation time
basic	6–9	30%–38%	Triangular(0.6, 0.9, 1.3) min/cup
fruit	10–13	35%–45%	Triangular(0.9, 1.3, 2.0) min/cup
topping	12–15	40%–50%	Triangular(1.0, 1.5, 2.3) min/cup
premium	16–18	45%–55%	Triangular(1.3, 2.0, 3.0) min/cup

Table A.3: Scenario Design (Core)

Scenario	Promotion	Staff	Role
A	No	2	Normal-operation baseline
B	Yes	2	Promotion only, without extra staffing
C	Yes	3	Promotion with one additional staff member
D	Yes	2	Promotion with no extra staff, but higher process efficiency

Table A.4: Key model parameters

Parameter	Meaning in this project
`lambda_per_hour`	Average number of orders arriving per hour. It represents demand intensity, and is higher in promotion scenarios to reflect increased customer traffic.
`staff`	Number of employees working during the peak period. It represents service capacity: more staff can reduce queue pressure, but also increases labour cost.
`promotion`	Indicates whether the second-cup-half-price promotion is applied. It separates normal operation from promotion scenarios and affects revenue calculation.
`order_size_r`	Controls the proportion of multi-cup orders. A higher value means customers are more likely to buy two or more cups, reflecting add-on purchases or group orders under promotion.
`product_mix`	Probability mix across basic, fruit, topping, and premium drinks. It affects average selling price, variable cost, and preparation complexity.

Parameter	Meaning in this project
`prep_multiplier`	Adjustment factor for preparation time. A lower value represents improved process efficiency, as used in Scenario D.
`fixed_order_time`	Fixed time required for ordering, payment, confirmation, and handoff communication. It captures the base service time for each order.
`patience`	Customer patience threshold before abandonment. It represents how long a customer is willing to wait before leaving the queue.
`batch_efficiency`	Efficiency gain from preparing multiple cups in the same order. It reflects shared preparation steps, so service time does not increase fully linearly with cup count.

B Outputs from the Order Generation Process

Table B.1: Outputs from the order generation process

Return Value	Meaning	Usage
`arrival_times`	Order arrival time for each order	Process orders chronologically in the queuing model
`order_sizes`	Number of cups per order	Calculate the average order size
`revenues`	Revenue generated per order	Included in total revenue after successful service
`variable_costs`	Raw material cost per order	Included in total cost after successful service
`service_times`	Service duration per order	Determines how long staff members are occupied

C Outputs and Indicators from the Queuing Model

Table C.1: Outputs and indicators from the queuing model

Return Field Name	Business Definition & Description
`profit`	Total profit generated in the simulation scenario (Total Revenue - Total Variable Cost - Total Labour Cost)
`revenue`	Total revenue from all successfully served orders during the peak period
`variable_cost`	Total raw material variable cost from all successfully served orders during the peak period
`labour_cost`	Total labour cost incurred during the peak period
`served_orders`	Total number of orders successfully served during the peak period
`lost_orders`	Total number of lost orders (customer abandonment/timeout while waiting) during the peak period
`total_arrivals`	Total number of orders arrived during the peak period (successfully served orders + lost orders)
`lost_potential_profit`	Total potential profit loss from lost orders
`avg_wait`	Average waiting time of all successfully served orders
`prob_wait_all`	Probability of waiting for all arrived orders [4]
`prob_wait_served`	Probability of waiting among successfully served orders
`prob_wait_over_8_all`	Probability of waiting longer than 8 minutes for all arrived orders [2]

Return Field Name	Business Definition & Description
prob_wait_over_8_served	Probability of waiting longer than 8 minutes among successfully served orders
abandonment_rate	Order abandonment rate or customer churn rate (Number of lost orders / Total number of arrived orders) [1]
avg_queue_length	Average queue length during the peak period
utilisation	Staff utilisation rate during the peak period (Total staff busy time during peak / Total staff available time – 180 minutes) [3]
workload_ratio	Workload ratio (Total service time of all successfully served orders / Total staff available time, including service time beyond the peak period – beyond 180 minutes) [5]
avg_order_size	Average number of cups per order

D Summary of Downside Risk Metrics

Table D.1: Summary of downside risk metrics.

Scenario	Description	VaR_{95} Shortfall (£)	$CVaR_{95}$ Shortfall (£)	Probability of Missing Target
A	No promotion, 2 staff	212.2	260.8	50.0%
B	Promotion, 2 staff	74.4	108.8	28.1%
C	Promotion, 3 staff	59.4	112.8	11.1%
D	Promotion, 2 staff + process improvement	0.0	47.7	5.0%

E Parameters Tested in the OAT Sensitivity Analysis

Table E.1: Parameters tested in the OAT sensitivity analysis.

Parameter Tested	**Low Setting**	**High Setting**	**Meaning**
Arrival rate (λ)	40 customers per hour	60 customers per hour	The range of expected customer arrival rates during promotions
Order-size parameter (r)	0.42	0.62	The probability that a customer will place a multi-cup order
Preparation time multiplier	0.85	1.15	A factor adjusting the base drink preparation time, faster or slower than average
Customer patience	(4, 6, 10) minutes	(6, 10, 14) minutes	The distribution of customer waiting tolerance before abandoning their order
Raw material cost multiplier	0.90	1.10	Fluctuations in the cost of raw materials, including ingredients, cups, lids, and other consumables

F Average Queue Length and Staff Utilisation Across Scenarios

Table F.1: Average Queue Length and Staff Utilisation Across Scenarios

Scenario	Average Queue Length	Staff Utilisation
A	0.95	74.49%
B	2.97	94.41%
C	0.94	74.32%
D	2.13	87.32%

G Downside Risk Metrics Across Scenarios

Table G.1: Downside Risk Metrics Across Scenarios

Scenario	VaR95 Shortfall	CVaR95 Shortfall
A	212.22	260.76
B	74.36	108.81
C	59.51	112.78
D	0.00	47.67